

AMAS: Problem set 3



Aske Roelshøj (KVF851)
Due Wed. 11th of March 2026

Problem 1

1a

We start by loading the data, dividing into high and low earners, and sample 500 high and low earners from our data to save as a test dataset - and remove these from the training data.

For this problem we'll use Sci-kit Learns' AdaBoostClassifier to classify the data. We then split our data into other parameters, i.e. age, education, gender and hours-per-week, and into earnings, i.e. the parameter we want to separate with respect to.

We then create our BDT with a max depth = 3, amount of estimators being 300 and a learning rate of 0.5. We tried to fiddle around with these metrics, but we couldn't really get an amazing separation.

Using Sci-kit Learns' Confusion matrix and Accuracy score packages on our test data, we get a classification accuracy of 64.70%.

On the same test data, we get the following histogram in fig. 1 along with a proposed separation threshold for the BDT score.

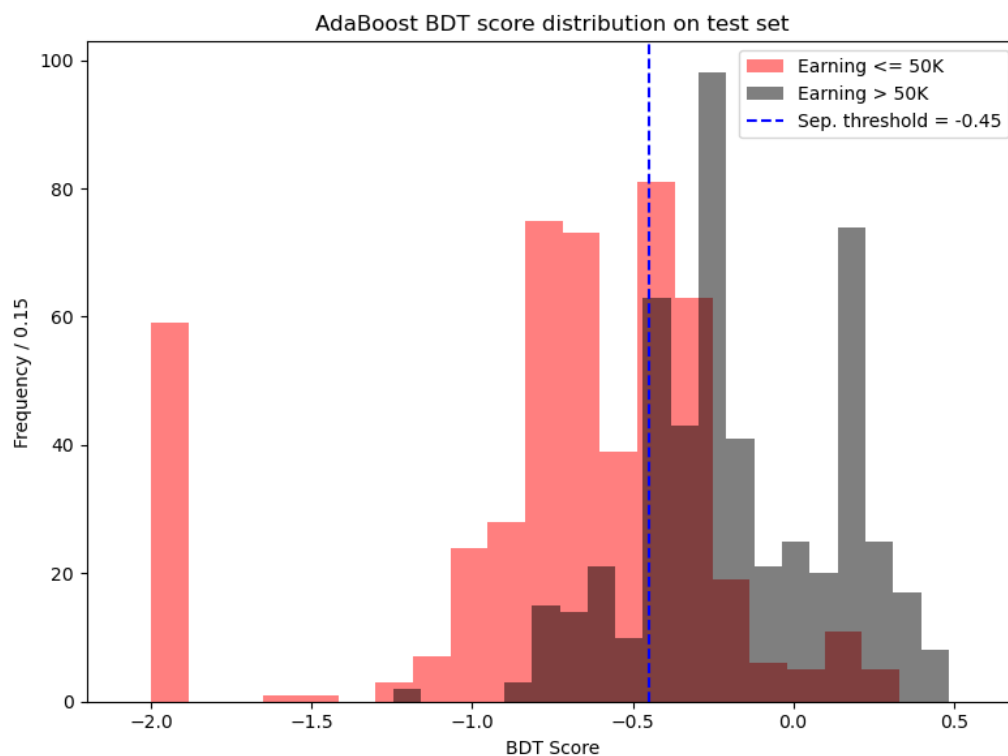


Figure 1: Histogram showcasing the BDT scores for low and high earners on test data.

1b

In the training set, 82.61% are low earners and 17.39% are high earners.

One pitfall for BDTs is that the model can be overtrained, such that it mistakes statistical fluctuations for real data features. We've tried to not do this by ensuring that the max depth and that the number of estimators are not too high, such that the model complexity is not

too high. Looking at slide 8 from Lect. 10 we see that prediction error can begin to grow if the model is too complex.

1c

Running our trained BDT on the real sample and using the proposed -0.45 separation threshold, we estimate 3426 high earners and 3086 low earners. The files containing the IDs for the high and low earners are uploaded.

Problem 2

2a

Looking at the dataset there are two columns for latitude and longitude, so we use the first for each metric and use the "backup" column to fill NaN-values.

Looking at google maps, we restrict latitude and longitude values to $[35.66, 35.38]$ and $[-78.93, -78.72]$ respectively, and plot the incidents:

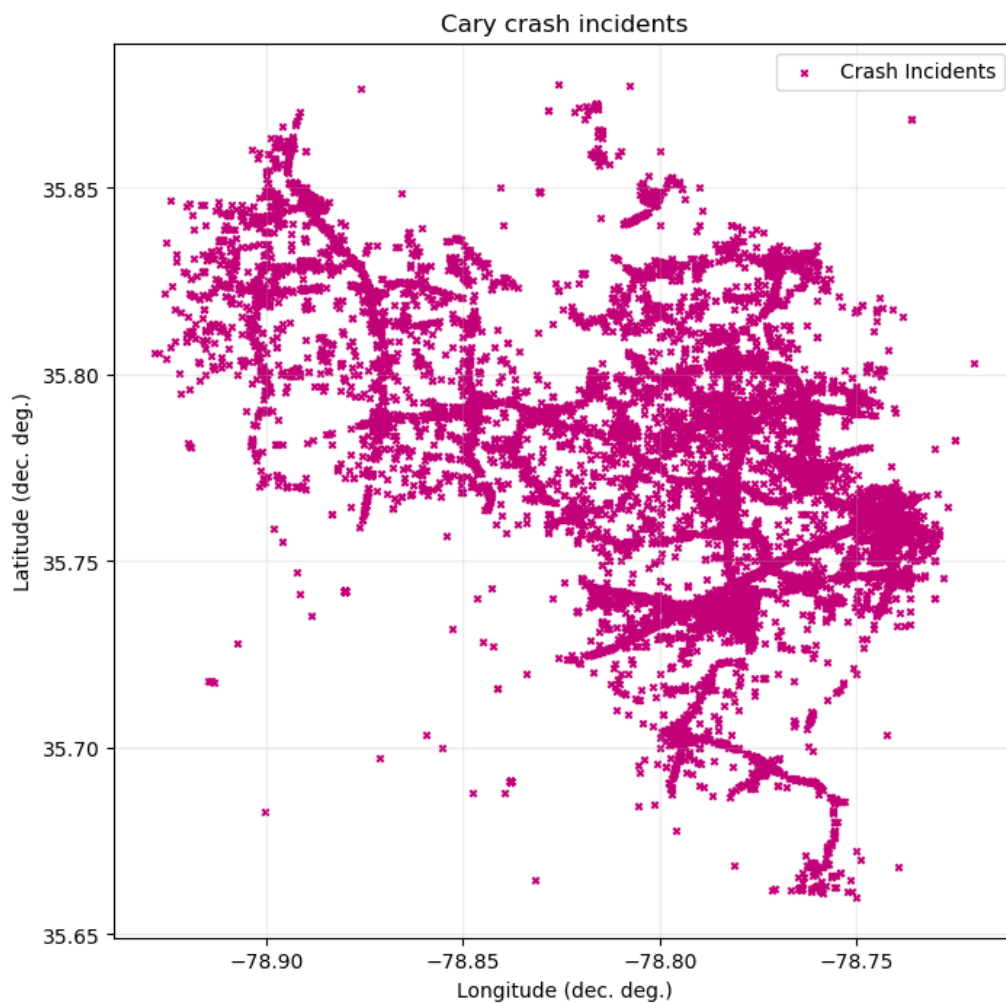


Figure 2: Scatterplot showcasing all crash incidents in Cary.

We also load the times columns and reformat the data to an hour format. Then we get the histogram shown on fig. 3.

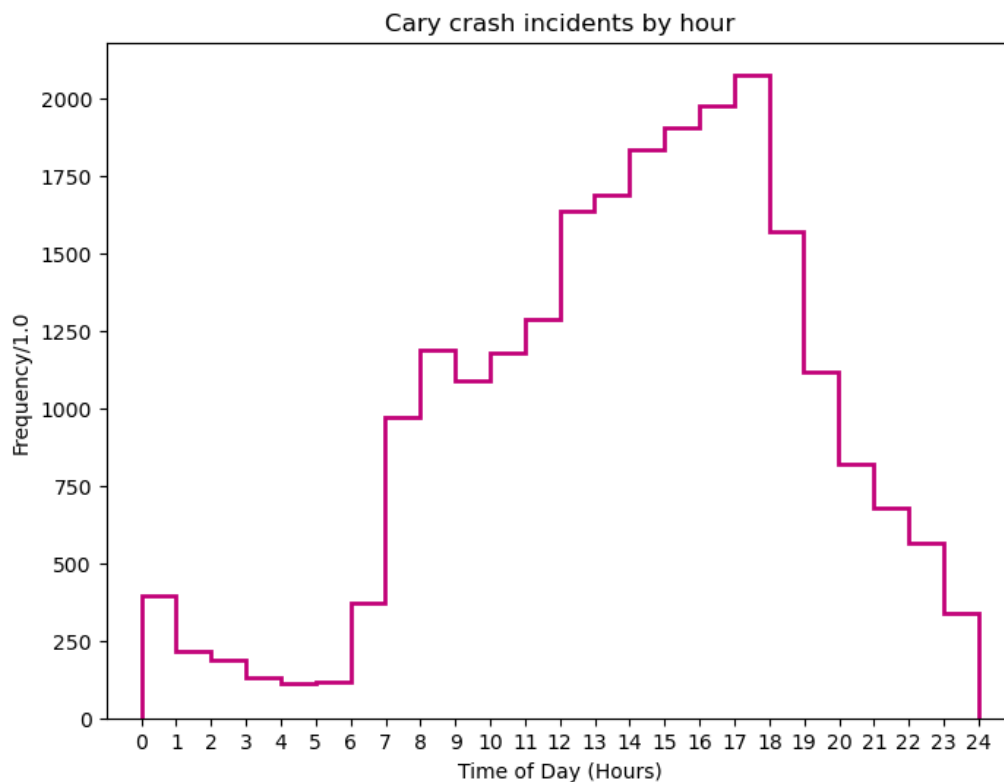


Figure 3: Histogram of crashes by hour in Cary.

To produce a KDE using a Gaussian kernel on the data with a bandwidth of 0.25, we would simply define the Gaussian kernel using the equation on slide 17 of Lect. 11 with a bandwidth of 0.25, and use it on the times data - and remember to normalize it! To handle the wrapping of time to correctly handle events close to each other in time, we would create tile the times, i.e. create an array 3x the size of the original array from -24 hours to 48 hours - and then also remember to divide by 3 to get the correct PDF values.

2b

We use the above recipe but now using a Epanechnikov kernel instead, with a bandwidth of 0.8 and using 10000 points. We draw the KDE on top of fig. 3, and obtain fig. 4.

We also evaluate the KDE at the specified times, and obtain the results in table 1.

Time	Time (in hours)	KDE value
00:23	0.3833	0.0132
01:49	1.8167	0.0089
08:12	8.2000	0.0498
15:55	15.9167	0.0806
18:02	18.0333	0.0790
21:12	21.2000	0.0307
23:44	23.7333	0.0101

Table 1: KDE values at specified times

To estimate what 2-hour window would be the best to add additional police officers, we, in small steps, integrate our KDE PDF in 2 hour windows and find which window yields the

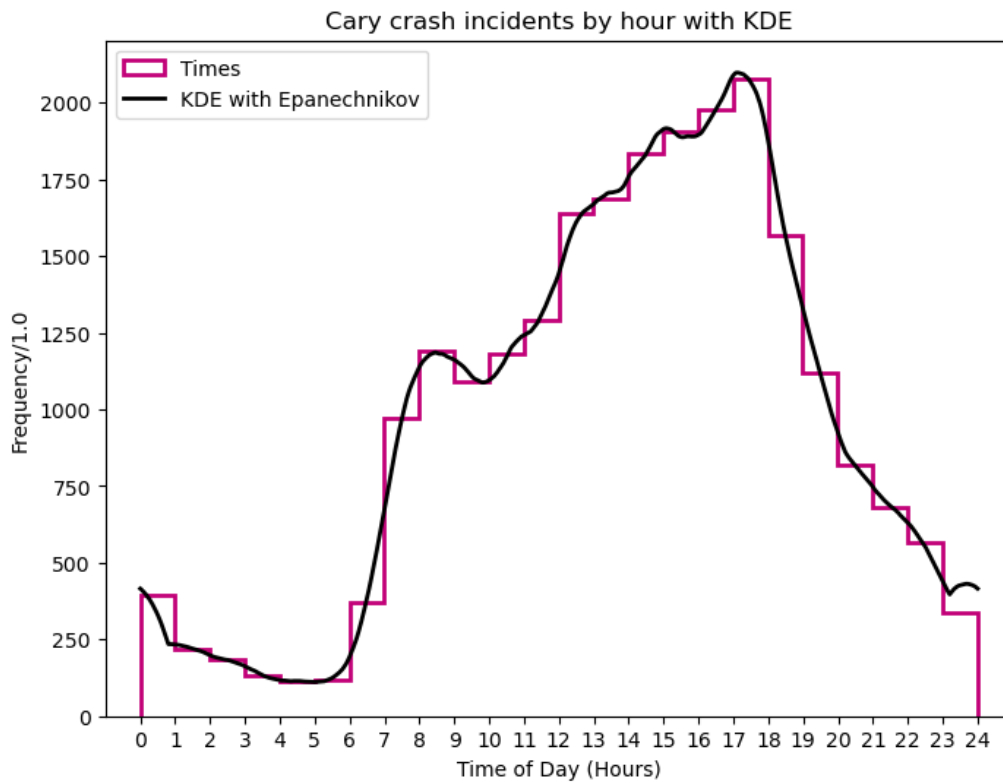


Figure 4: Histogram of crash incidents by hour with KDE using Epanechnikov kernel drawn on top.

greatest value, i.e. we find which 2 hour window results in the most crashes. We do this by finding the amount of bins in a 2 hour interval, and the use numpys cumsum to determine the integrals.

We find the that the best two-hour window to patrol is from 15:57 to 17:57.

We then find the amounts of crashes in that window and reduce it by 10%. We then find the new total amounts of crashes after this reduction, and compare it to the total amount without any reduction.

Percentage wise, we get a 1.71% reduction in the 24 hour crashes if there are additional patrols from 15:57 to 17:57.

2c

To do Epanechnikov KDE in 2d, we simply make a grid to scan over and for each point determine u_x, u_y , and if $|u_x|, |u_y| \leq 1$, then we assign that point a value which is the product of the x and y kernel.

We make the grid 500x500, and choose the same range for latitude and longitude as in fig. 2. We get then get fig. 5.

To find the percentage of crashes in longitude $[-78.76, -78.72]$ and latitude $[35.74, 35.78]$, we simply divide the sum of KDE in that box by the total KDE.

We get that the percentage of crashes in longitude $[-78.76, -78.72]$ and latitude $[35.74, 35.78]$ is 18.62%.

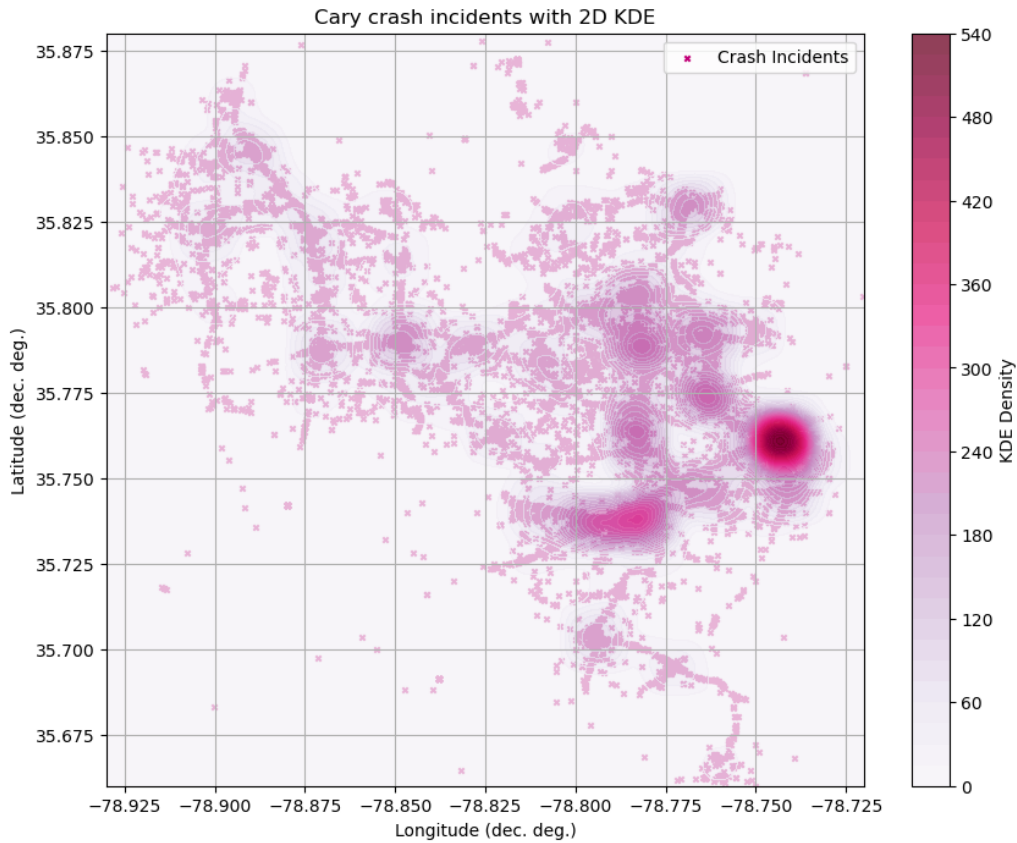


Figure 5: 2D KDE with Epanechnikov kernel for crash incidents in Cary.

Problem 3

3a

To determine $P(A_2|D)$, we'll use Bayes theorem to calculate the posterior:

$$P(A_2|D) = \frac{P(D|A_2)P(A_2)}{P(D)} = \frac{P(D|A_2)P(A_2)}{\sum_i P(A_i)P(D|A_i)} = \frac{0.04 \cdot 0.15}{0.03275} = 0.1832 \approx 18.32\% \quad (1)$$

So the probability that the defective pacemaker came from A_2 is 18.32%.

Doing the same calculation as eq. (1), we get the following table:

Facilities	Prior	Likelihood	Posterior
A1	0.3500	0.0200	0.2137
A2	0.1500	0.0400	0.1832
A3	0.0500	0.1000	0.1527
A4	0.2000	0.0350	0.2137
A5	0.2500	0.0310	0.2366

Table 2: Posterior probabilities for each facility

From table 2, it is evident that it is most probable that a defective pacemaker comes from A5 with a probability of 23.66%.

3b

If we want $P(A_j|D) = 1/5, \forall j$ and want the priors to stay the same, we end up with:

$$P(A_j|D) = \frac{P(D|A_j)P(A_j)}{\sum_i P(A_i)P(D|A_i)} \quad (2)$$

if $P(D|A_j)P(A_j) = c, \forall j$, then:

$$P(A_j|D) = \frac{P(D|A_j)P(A_j)}{\sum_i P(A_i)P(D|A_i)} = \frac{c}{5c} = 1/5 \quad (3)$$

and if $P(A_j)$ must be constant, then:

$$P(D|A_j)_{new} = c/P(A_j) \quad (4)$$

Since $P(A_j)$ are all below 1, we want the smallest c . If the defect rates only can increase, we have:

$$P(D|A_j)_{new} = c/P(A_j) \geq P(D|A_j)_{old} \Rightarrow c = \max_j (P(A_j)P(D|A_j)) \quad (5)$$

since $c \geq P(A_j)P(D|A_j), \forall j$ thus the best choice is the smallest c possible.

Doing this, we get $c = 0.00775$ and results on table 3.

Facility	Original $P(D A_i)$	New $P(D A_i)$	New Posterior
A1	0.0200	0.0221	0.2000
A2	0.0400	0.0517	0.2000
A3	0.1000	0.1550	0.2000
A4	0.0350	0.0387	0.2000
A5	0.0310	0.0310	0.2000

Table 3: New defect rates for equal posteriors

Problem 4

4a

We start by loading the data. To solve this problem we create a Metropolis-Hastings MCMC algorithm, using $P(x)$ as our proposal function. We implement the skipping valley caveat by only stepping in integer values, and also implement periodic boundary conditions using modulo - and then implement Metropolis-Hastings.

For each beta value we calculate the analytic estimate and create 1000, 100000 and 1000000 samples, we get fig. 6.

The convergence of the model is VERY sensitive to what value of β we choose. We've tried to run the simulation in fig. 6 numerous times and a lot of the times the last plot is completely off the analytic estimate. This is probably due to the fact that the chain become more

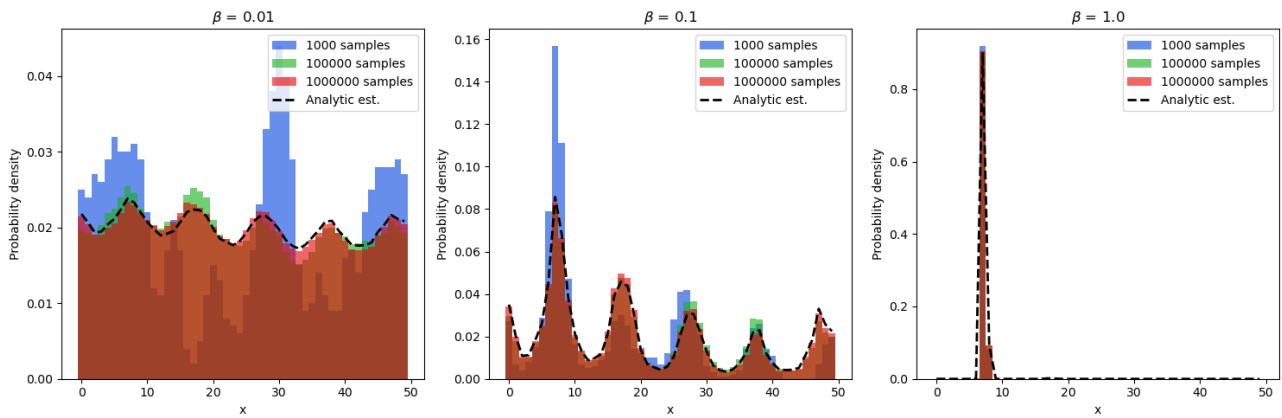


Figure 6: For different beta values, histogram showcasing convergence of distributions as the sample size increases.

and more narrow around a spot when β increases, so it becomes very sensitive to starting guesses. The algorithm used uses a random starting position. The problem might not be suited for MCMC - i.e. we are not sure if the problem fulfills the Markov assumptions.

We can create a timestep distribution by saving the amount of stops/dwells in a position. We get fig. 7.

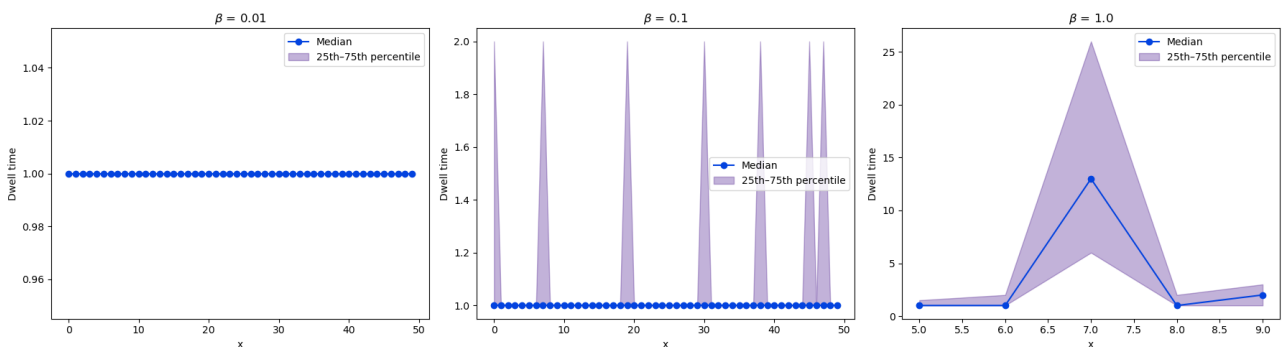


Figure 7: Figures showcasing time step distributions for different β values.

It doesn't really look correct, but it's the best we got...

4b

Now we use $\beta = 0.25$ and determine the rate, τ , to be $\tau = 1/25$, since we want 25 steps to occur before attaching.

We modify our MH algorithm so that there is an attachment boolean, and that each iteration there is a τ chance of attaching.

Obtaining the same amount of samples as in fig. 6 to compare and adding the analytic solution, we get fig. 8.

We actually see a pretty big disagreement... Lets try and do it for different τ , maybe $1/100$, $1/25$ and $1/10$. We then get fig. 9.

We can also add a left or right bias using $\tau = 0.04$, and compare in fig. 10.

As can be seen biasing shifts the distribution in the biased direction (sorry for bad subplot names i ran out of time...).

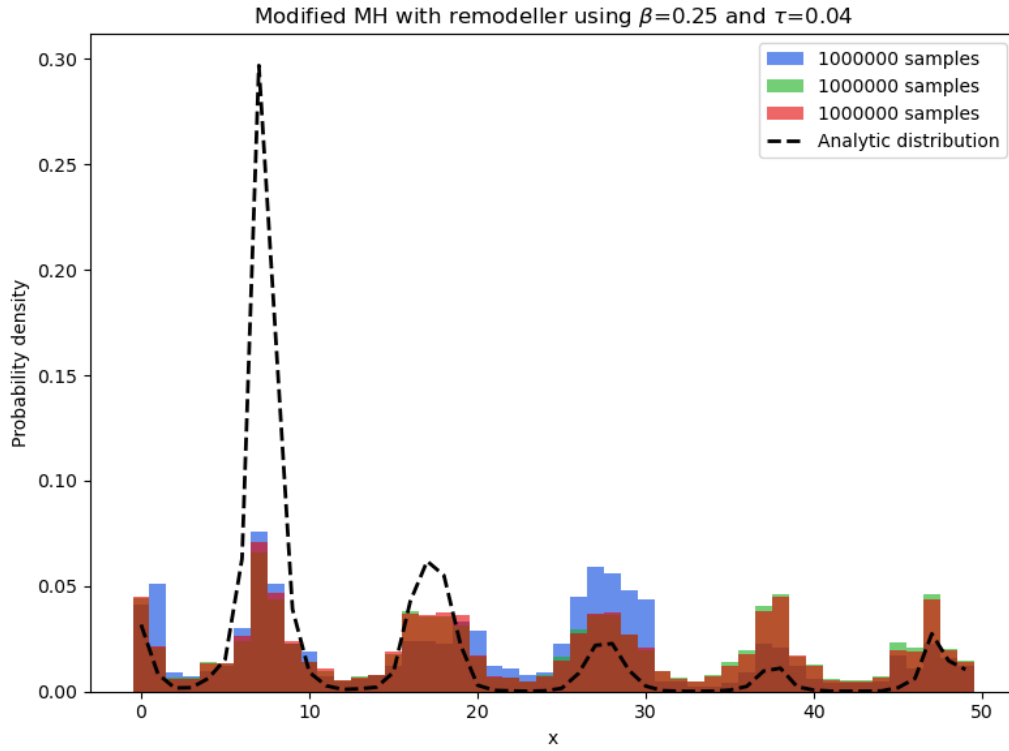


Figure 8: MH model with remodeller setting $\beta = 0.25$ and $\tau = 0.04$.

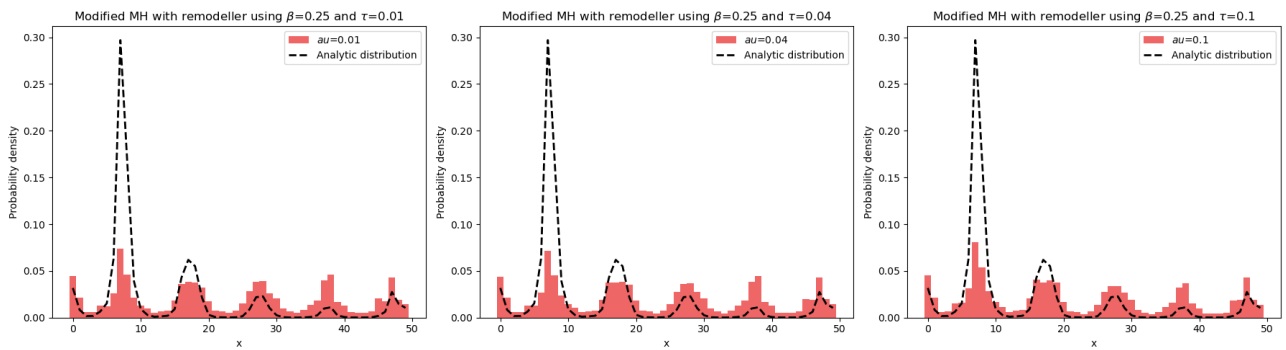


Figure 9: MH with remodeller for $\tau = 0.01, 0.04, 0.1$. (sorry for legend name wrong, its supposed to be τ but i dont have time to fix...)

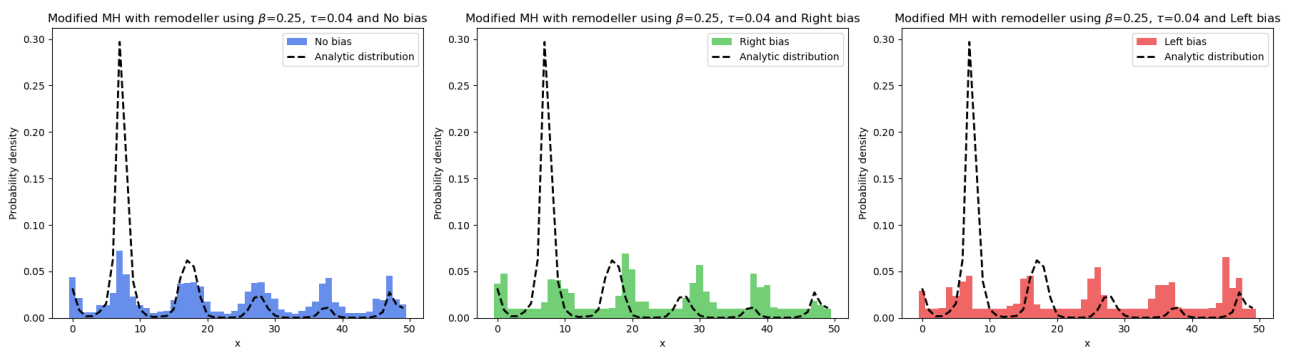


Figure 10: MH with remodelling with No, Right and Left bias.